

MUKESH KUMAR SHARMA

Senior / Lead iOS Engineer

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SUMMARY

Senior iOS engineer with 14+ years of deep expertise in the Apple ecosystem, delivering scalable, high-performing, AI-powered mobile applications from concept to App Store launch. Passionate about performance, reliability, and user delight as foundational principles of mobile development. Deep expertise in Swift, SwiftUI, Combine, and Swift Concurrency (async/await, Actors, Sendable), with a proven track record of integrating cutting-edge iOS platform capabilities and enabling other developers through reusable frameworks and architectural best practices. Full-stack AI/mobile ownership through Doctor AI — a solo-built iOS + Node.js/TypeScript backend using OpenAI SDK, LLM prompt engineering, and Firebase — plus early open-source Swift integrations for OpenAI (chatGPT_iOS, 10★) and Stable Diffusion (StableDiffusion_iOS, 9★). Proven record architecting MVVM/Clean modules, establishing robust testing frameworks (80% coverage), owning CI/CD, and collaborating cross-functionally with product, design, research, and engineering teams. Based in London/Toronto, ON; Canadian Citizen open to US relocation on TN visa.

TECHNICAL SKILLS

Languages: Swift 5.x, SwiftUI, Objective-C, JavaScript/TypeScript (React Native), C/C++, Python

Cross-Platform Mobile: React Native (iOS + Android, production, 3+ years), native module bridging (Swift ↔ JS), Ionic 4 / Capacitor, Angular

iOS Frameworks: UIKit (programmatic and storyboard), Auto Layout, SwiftUI, Combine, Swift Concurrency (async/await, Actors, Sendable), Core Data, Core Animation, Core Graphics, AppIntents, Speech, AVFoundation, MapKit, WidgetKit

Architecture: MVVM, MVC, VIP/Clean, SOLID, Protocol-Oriented Programming, Dependency Injection, Modularization (SPM), Apple Human Interface Guidelines (HIG)

Testing: XCTest, XCUITest, TDD, mocks/stubs, Accessibility Inspector (VoiceOver)

Process & Collaboration: Agile / Scrum, sprint planning and estimation, cross-functional team collaboration (product, design, backend, QA), technical documentation, risk assessment and mitigation, JIRA, Confluence, Bitbucket

Networking & APIs: REST/JSON, GraphQL (Apollo iOS), OAuth 2.0, SSL/certificate pinning, WebSockets, URLSession, Codable

Backend & Tooling: Node.js + TypeScript (REST APIs, LLM orchestration, deployed on Google App Engine), Firebase (Auth, Firestore, FCM), Fastlane (Ruby), GitHub Actions, Xcode Cloud, Bitbucket, JIRA, Confluence, shell/Bash, Python

Performance & Quality: Instruments (Time Profiler, Allocations, Leaks), Crashlytics, Sentry, symbolication, launch-time and memory optimization

Security & Platform: Keychain, LocalAuthentication (Face ID / Touch ID), biometric auth, App Transport Security, Universal Links, Push (APNs), Notification Service Extensions

Experimentation: Firebase Remote Config, feature flags, A/B testing, Google Analytics, Grafana observability

Integrations: Bluetooth / MFi accessories, OpenAI SDK, Twilio Video, Apple Speech, MapKit, Core Location

AI / LLM on iOS: OpenAI SDK integration, prompt engineering, LangChain-based iOS prompt templates, on-device Speech-to-Text, AI-assisted development (GitHub Copilot, Claude, ChatGPT for scaffolding, refactoring, and test generation)

PROFESSIONAL EXPERIENCE

Lead iOS Engineer — Tech Mahindra (TruPOS — Enterprise POS App)

Jan 2025 – Present

- Led 0→1 development of TruPOS, the enterprise's central payment platform iOS app, in Swift/SwiftUI/Combine. The app serves as the single checkout hub for multiple business lines and external systems (e.g., Field Service 365), connected via deep links. Owned architecture (MVVM), payment checkout, transaction history, and merchant onboarding flows.
- Designed an offline transaction queue using Core Data + BGTaskScheduler with a 72-hour on-device data retention window, enabling field sales teams to process payments in remote areas or low-connectivity environments and automatically sync when connectivity returns — reducing transaction failures to effectively zero in offline scenarios.
- Built a reusable Swift Package for payment flows, receipt generation, and deep linking — adopted across 3 internal apps, eliminating the need to reimplement these modules per app.
- Implemented real-time payment status updates with async/await, Actors, and Sendable, eliminating data races and improving UI responsiveness during high-volume checkout.
- Integrated Bluetooth card readers and external accessories with full background-mode support; authored custom retry/error-handling layer for intermittent POS connectivity.
- Achieved 80% unit-test coverage on the business-logic layer using XCTest/XCTest; built reusable mocks, stubs, and test-data builders.
- Owned CI/CD via Fastlane, GitHub Actions, and Xcode Cloud — automated TestFlight and App Store releases, replacing a manual release process and substantially shortening release cycle time. Wrote internal Node.js tooling and mock servers to unblock parallel backend/iOS development.
- Profiled launch time and memory with Instruments (Time Profiler, Allocations, Leaks); diagnosed and fixed retain cycles and redundant work, measurably improving cold-start time and reducing memory pressure.
- Troubleshoot and debugged complex production issues including intermittent POS connectivity failures, payment-flow edge cases, and cross-device Bluetooth pairing conflicts — providing timely resolutions that maintained merchant uptime during peak transaction hours.
- Actively participated in Agile sprint planning, story-point estimation, and backlog grooming with product owners and cross-functional stakeholders; drove documentation on data requirements and risk mitigation for the payment application.
- Integrated Face ID / Touch ID via LocalAuthentication and Keychain for secure merchant login; wired Crashlytics + Sentry for symbolicated crash reporting and alerting.
- Drove VoiceOver accessibility across checkout and history screens following Apple's Human Interface Guidelines; partnered with designers to provide constructive feedback on iOS-native interaction patterns. Collaborated closely with backend engineers to integrate RESTful APIs and ensure seamless end-to-end functionality across payment, authentication, and observability flows.

Senior / Lead iOS Engineer — Tech Mahindra — NEOM KSA Smart City App

Jun 2023 – Jan 2025

- Architected and shipped the first production release of the NEOM Smart City iOS app to the App Store, supporting residents, visitors, and service providers.
- Built MVVM SwiftUI modules with async/await; designed scalable, reusable components (custom UINavigationController, ViewModifiers, ViewBuilders) adopted across multiple NEOM apps.
- Defined the architectural pattern, interface contracts, and versioning strategy for NEOMCore — a reusable SPM framework for networking, OAuth, SSL pinning, and shared UI — enabling developers across multiple teams by providing scalable, maintainable client infrastructure adopted across ~20 internal NEOM mobile apps. Integrated Apollo iOS for type-safe GraphQL queries and subscriptions against the NEOM services backend.
- Integrated OpenAI iOS client and a LangChain-based prompt SDK to power an in-app digital assistant for utilities, transport, and support workflows.
- Implemented Twilio audio/video calling for resident-to-agent support and Apple Speech framework for voice-driven interactions.
- Led a small iOS team over the 19-month NEOM project lifecycle: ran design-review sessions with cross-functional teams (product owners, designers, backend, security), owned architectural decisions and interface contracts, walked stakeholders through technical designs, mentored 3 engineers, and consistently delivered sprint commitments through to the production App Store launch.

- Built CI/CD with Fastlane, Ruby, Bash, and GitHub Actions; monitored production health with Grafana dashboards (API latency, crash-free rate).

Senior Mobile Engineer (React Native) — Tech Mahindra — TruGreen Mobile App (iOS + Android) *Mar 2023 – Dec 2024*

- Delivered production features for the TruGreen consumer mobile app in React Native (TypeScript), shipping to both iOS and Android from a single codebase.
- Authored native iOS modules in Swift and bridged them to the React Native JS layer — exposing platform capabilities (biometrics, deep links, secure storage) that pure JS could not reach.
- Debugged cross-platform issues across Metro, Hermes, and the native bridge; diagnosed production crashes using symbolicated Crashlytics for both iOS and Android builds.
- Collaborated with Android engineers and backend teams to align API contracts, shared component APIs, and release cadence across both stores.

Senior / Lead iOS & Release Engineer — Tech Mahindra — Kroger & Banner Apps *Jan 2023 – Jun 2023*

- Owned end-to-end iOS delivery for the Kroger Gamification module (v1.0 and v2.0), shipped to the Kroger iOS app user base across Kroger and sister-banner apps (Ralph's, Fred Meyer, Harris Teeter, and others).
- Authored and maintained DevOps pipelines (Fastlane, GitHub Actions, shell/Ruby/Node.js) used by the entire iOS org for build, sign, and deploy.
- Instrumented Gamification with Firebase Remote Config feature flags and A/B experiments; monitored rollouts via Crashlytics crash-free rate and Grafana dashboards.
- Wrote XCTest/XCUITest coverage for business-logic and integration paths; verified VoiceOver behavior with Accessibility Inspector.
- Conducted code reviews across the iOS team and mentored junior engineers on Swift/SwiftUI best practices.

Senior iOS Engineer — Avenue Code @ GAP Inc. — GAP Browse App *Jun 2022 – Dec 2022*

- Delivered iOS features across GAP, Athleta, Banana Republic, and Old Navy storefronts in Swift/SwiftUI and Objective-C.
- Led the Notification System workstream from scratch through production launch; owned story-pointing, code reviews, and stakeholder demos.

Senior iOS Engineer — Clearbridge Mobile / Amdocs — TD Canada Banking App *Sep 2021 – Jun 2022*

- Led development on SIJ, MHA, and CIBA loan modules in Swift and Objective-C, shipping features used by millions of TD banking customers.
- Integrated Face ID / Touch ID via LocalAuthentication and Keychain for secure customer authentication flows; hardened the app against jailbreak and tampering.
- Contributed to the TD iOS Banking Foundation Framework and republished it via CocoaPods to streamline dependency management.
- Mentored junior engineers and co-ops; partnered with product and business stakeholders on requirements and demos.

Lead iOS Engineer — Tata Consultancy Services — Trickl Utility & Green Button *Nov 2017 – Aug 2021*

- Owned the iOS codebase for Trickl, a smart-utility app: built real-time energy gauges, device-control toggles, and push notifications; migrated codebase from Swift 3 → Swift 4.
- Architected Core Data persistence and a GCD-based networking layer integrating with AWS backend services.
- Built cross-platform Green Button utility apps (Toronto Hydro, London Hydro, Whitby Hydro) using Ionic 4/Capacitor + Angular 9 + TypeScript, sharing reusable iOS components as a framework.

iOS Engineer — Tata Consultancy Services — Pfizer (Events, Emotions, GroAssist, Gavi Stats, KC Planner) *Jan 2015 – Nov 2017*

- Shipped six internal Pfizer iOS apps in Objective-C and Swift: built reusable expandable table cells, custom Core Graphics charts (scatter, bar, rocker), Janrain SSO integrations, and offline sync modules.

- Authored technical design docs, class/sequence diagrams, and provided estimates; mentored teammates and ran code-quality reviews.

iOS Engineer — GlobalLogic, smartData, Regard Solutions

Aug 2011 – Dec 2014

- Shipped 8+ iOS apps to the App Store across healthcare, retail, and consumer domains (Reliance Jio World, Wall Mall, MindBody Yoga/Fitness, Digital PA, Emergency Link, Share My Pair, EQY-P, Viva iPad).
- Worked across Objective-C, Titanium, and early iOS SDKs (iOS 5–7); contributed to accessibility R&D for the Qualcomm Healthy Circle POC.

PERSONAL PROJECTS

Doctor AI — iOS + Android App + Node.js Backend (TestFlight Beta) — Personal Project

Jan 2024 –

Present

- Designed, built, and shipped Doctor AI end-to-end as a solo full-stack project spanning iOS, Android, a shared Swift Package, and a Node.js backend. Live TestFlight beta used by external testers.
- iOS client in Swift/SwiftUI: integrated Apple's Speech framework for real-time transcription and Twilio Video for secure doctor–patient calls. Shared domain logic via a dedicated Swift Package (AIDoctor-SPM).
- Built a Node.js + TypeScript backend deployed on Google App Engine: REST endpoints, auth, session management, and LLM orchestration with prompt engineering to generate consultation notes and surface potential diagnoses.
- Delivered an Android companion app and shared AI-component prototypes (Jupyter Notebook) for model evaluation and iteration.
- Used Firebase for authentication, realtime data, and push notifications; wired crash reporting and basic analytics for the beta cohort.
- Full ownership of the stack: architecture, API contracts, schema design, deployment, iOS ↔ backend integration, and TestFlight release management across 6+ repositories.
- Tech: Swift, SwiftUI, Combine, Apple Speech, Twilio Video, TypeScript, Node.js, Google App Engine, Firebase, OpenAI / LLM prompt engineering.

Open-Source iOS + AI Experiments — github.com/imuks

2023 – Present

- chatGPT_iOS (Swift, MIT, 10★) — native iOS client for OpenAI's chat API; one of the earlier public Swift integrations (Jan 2023).
- StableDiffusion_iOS (Swift, MIT, 9★) — on-device Stability AI / Stable Diffusion integration for iOS image generation.
- swift_DSA — Data Structures and Algorithms implemented in Swift; used for interview prep and as a public reference.
- Pharma-iOS, MedPalAI-iOS, reactiv-ios — additional Swift/SwiftUI projects exploring healthcare AI, reactive architecture, and interview patterns.

EDUCATION

B.E., Computer Technology — Priyadarshini College of Engineering, R.T.M. Nagpur University, 2011

ARCHITECTURAL LEADERSHIP & CROSS-TEAM IMPACT

- Solution design & documentation: translate business requirements into documented technical designs — architecture diagrams, interface contracts, data requirements, integration points, infrastructure specs, and risk mitigation plans — across TruPOS, NEOM, Kroger, and TD Banking engagements.
- Reusable frameworks adopted across teams: NEOMCore SPM (~20 NEOM apps), TruPOS payment SPM (3 internal apps), TD iOS Banking Foundation Framework (redeployed via CocoaPods), and Green Button cross-platform components (multiple Ontario hydro utilities).
- Stakeholder & design-review leadership: drove design-review sessions and walkthroughs with product, backend, and security stakeholders to align iOS architecture with broader technology roadmaps and ensure consistency across platform apps.

- Mentorship & knowledge transfer: mentored iOS engineers across NEOM, Kroger, GAP, and TD Canada engagements through code reviews, architecture sessions, and effective problem solving; authored technical design docs to support onboarding and continuity.

SELECTED HIGHLIGHTS

- 14+ years of iOS delivery, including 10+ years in Swift and 5+ years in SwiftUI/Combine.
- Shipped 30+ production iOS apps across enterprise, retail, banking, healthcare, and smart-city domains.
- Active open-source presence: early Swift integrations for OpenAI (chatGPT_iOS, 10★) and Stable Diffusion (StableDiffusion_iOS, 9★), plus a public Swift DSA repository.
- Deep CI/CD ownership: Fastlane, GitHub Actions, Xcode Cloud, shell/Ruby/Node.js automation.
- Track record mentoring engineers, leading small teams, and partnering with product and backend on ambiguous, high-stakes launches.
- Drives technical innovation in AI-powered mobile applications — adopted Swift Concurrency, SwiftUI, AppIntents, OpenAI SDK integration, and on-device ML tooling ahead of industry curve; passionate about expanding the boundaries of what iOS apps can do with AI.
- Proactively uses AI tools (GitHub Copilot, Claude, ChatGPT) in daily development workflow for code scaffolding, refactoring, and test generation — while critically reviewing all AI-generated output for quality, performance, and correctness.